

GUJARAT UNIVERSITY

Department of Computer Science

Master of Computer Applications (MCA) - Semester III

Subject: Computer Vision

Theory Assignment - 1

1. What is Computer Vision? Explain its scope and its relationship with Image Processing and Artificial Intelligence.
2. Give an overview of Computer Vision applications in domains such as medical imaging, robotics, surveillance, and autonomous vehicles.
3. Explain the Image Formation Model with respect to a Monocular Imaging System.
4. Explain Radiosity and its role in the physics of image formation.
5. Define Radiance and Irradiance and explain their relationship in the context of image formation.
6. What is BRDF (Bidirectional Reflectance Distribution Function)? Explain how it is used to model the reflectance properties of surfaces.
7. Explain how color is formed and represented in an image, and discuss its importance in Computer Vision.
8. Differentiate between Orthographic Projection and Perspective Projection with the help of diagrams.
9. Explain the Camera Model and describe its intrinsic and extrinsic parameters.
10. What is Camera Calibration? Explain its purpose and describe a general calibration procedure.
11. Explain Binocular Imaging Systems and their role in depth perception.
12. What is Multiple View Geometry? Explain its significance in 3D scene reconstruction.
13. Explain the concept of Structure Determination in Computer Vision.
14. What is Shape from Shading? Explain its assumptions and the steps involved in recovering shape information from an image.
15. Explain Photometric Stereo and Depth from Defocus techniques, and describe how a 3D model can be constructed from a set of images.